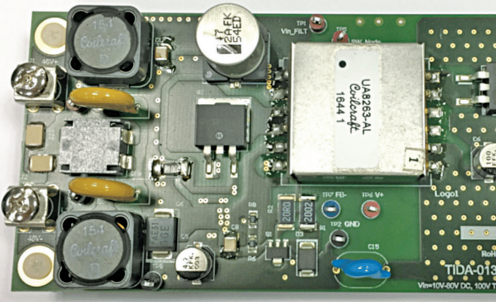
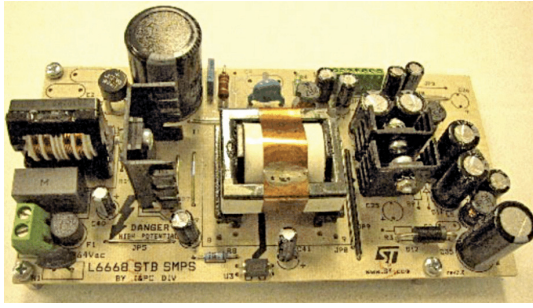


Interesting Reference Designs Of FLYBACK CONVERTERS

Title of Reference Design	Flyback 48V Car Battery	Flyback Converter For Set-Top Boxes
Image		
Key Attractions	The reference design provides a 12V power supply with a wide input range, surge protection, low noise, and primary-side regulation for use in automotive systems and harsh environments.	The reference design offers efficiency, low standby power use, wide input voltage support, multiple outputs, built-in protection, and compliance with safety and EMI standards for use in digital devices.
Highlights	The TIDA-01344 is a power supply design from Texas Instruments made for cars that use 48-volt batteries, like hybrid and electric vehicles. It changes the 48 volts into a steady 12-volt output, up to 2 amps, and can handle a wide input range from 10 to 80 volts. It is built to survive sudden spikes up to 100 volts and includes filters to reduce electrical noise, meeting car industry standards. The design uses a special type of circuit called a flyback converter that doesn't need extra parts like an optocoupler, making it smaller and more reliable. It also includes safety features to protect against power surges. The system works well alongside other converters to ensure 12-volt power is always available, even if one part fails. A special chip, called the LM5022-Q1, controls everything and helps the system respond quickly to changes. This design is useful in many parts of electric and hybrid cars, like motor systems, transmissions, and control units.	The reference design from STMicroelectronics is optimised for set-top box applications and supports a wide input voltage range of 90-265Vrms at 45-66Hz, delivering four distinct outputs with an average power output of 40W and a peak of 51W. It features the advanced L6668 controller, operating in fixed-frequency PWM current mode, and is engineered for high efficiency, consuming less than 1W in standby with a residual 5V at 50mA, aligning with global standby regulations. The design emphasises cost-effectiveness and simplicity through a low component count, while offering key protections such as short circuit protection with automatic restart. It meets EN50022-Class B standards for electromagnetic interference and complies with EN60065 safety standards, maintaining a creepage and clearance distance of 6.4mm.
Applications	The reference design is suitable for car motors, gears, controls, and battery systems.	The reference design is suitable for applications like HDD, SCART, and LNBP21 for satellite STBs.
OEM Brand	Texas Instruments (TI)	STMicroelectronics
URL	https://www.electronicsforu.com/electronics-projects/flyback-reference-design-for-48-v-car-battery	https://www.electronicsforu.com/electronics-projects/reference-design-of-flyback-converter-for-set-top-boxes